

Solving Linear Equations

Example

Expand and simplify: $2x - 6 = 8$

Step 1

Ask yourself: "What is the question asking me?"

It is asking us to find out the value of x . Something multiplied by 2 and then subtract 6 is equal to 8.

Step 2

- Your aim is to gather the letters on one side and numbers on the other.
- You are only able to cancel out one term or number at a time.
- You must always use the inverse when cancelling a term.

Step 3

Start by cancelling out the 6. To do this, you will need to 'do the opposite to it' (apply its inverse).

+6	$2x - 6 = 8$	+6
	$2x = 14$	

Step 4

Now you can separate the 2 from the x . Again, to do this, you will need to apply its inverse.

÷2	$2x = 14$	÷2
	$x = 7$	

Your final answer is $x = 7$

